

Johnson Jasson

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EDUCATION

Alfred University

Alfred, NY

Bachelor of Science in Electrical and Electronics Engineering (Mathematics Double major)

Aug. 2025 – May 2029

- First-Generation Student — Presidential Scholar — GPA: 4.0/4.0

Break Through tech

Artificial Intelligence (AI), Machine Learning (ML)

March. 2026 – present

CodePath Technical Interview Prep

Python, Data Structures & Algorithms

Sep. 2025 – Dec. 2025

McKinsey & Company Forward Program

Problem-Solving, Leadership, and Digital Transformation

Sep. 2025 – Dec. 2025

- Developing practical skills in structured problem-solving, leadership, and digital transformation.

TECHNICAL EXPERIENCE

AI-Powered Digital Twin for Battery Management in Renewable Microgrids

Alfred, NY

Data analysis, Machine Learning, Python, AI-coding

Aug. 2025 – Dec. 2025

- Developed an AI-driven digital twin using Convolutional Neural Networks (CNNs) to predict battery State of Health (SOH) and State of Charge (SOC).
- Addressed microgrid instability by integrating self-learning Battery Management Systems (BMS) for renewable energy optimization; Battery efficiency improved by 30%.
- Presented findings to faculty, researchers, and industry experts from GE Vernova, EDIBON, NYSEG and EPRI during the AU Energy Conference 2025.

Wi-Fi Smart Switch Using ESP32

Alfred, NY

Arduino, ESP32 and Breadboard prototyping

Dec. 2025

- Designed and built an ESP32-based Wi-Fi smart switch enabling remote ON/OFF control via a mobile application.
- Implemented embedded firmware in C/C++ (Arduino framework) to control GPIO outputs via an HTTP server.
- Debugged and validated hardware wiring, USB drivers, serial communication, and network connectivity during development.

RESEARCH AND PROJECTS

GE Aerospace Explore Electrical Engineering Job Simulation | Forge

Nov. 2025 – Dec. 2025

- Acquired technical proficiency in electrical distribution planning and design using AutoCAD, circuit breaker selection, and compliance with FAA and EASA Aerospace standards using industry-standard tools and methodologies.
- Developed critical thinking and problem-solving abilities to diagnose and strategize solutions for avionics reliability in challenging environmental conditions, employing simulation software for environmental impact assessment.

LEADERSHIP AND INVOLVEMENT

President NSBE Chapter, Alfred University

Alfred, NY

Leadership, Communication, Team work

Jan. 2026 - Present

- Launched and lead the campus chapter, increasing awareness of NSBE and creating a supportive platform for Black engineering students to showcase and develop their skills.

TECHNICAL SKILLS

Circuit Design & Testing: Analog/Digital Circuits, Breadboarding, Debugging, Electrical Circuit Analysis

Software: MATLAB, AutoCAD, GIS, SOLIDWORKS, MS Excel, LaTeX, Power Point.

Programming: Python, C/C++, Google Colab, Verilog, VHDL

Instrumentation: Oscilloscope, Ohmmeter, Clamp Meter, DMM, Soldering, Function Generator